

2-D Kinematics

SS ILP Physics (Intermediate), Week 1 Class 3



Outline

- Kinematics with constant acceleration (continued)
- Projectile Motion
- 2-D Kinematics
- Projectile Motion: Zero launch angle

Another equation of motion with constant acceleration

- For constant acceleration, acceleration along the x -axis = a_x
- Velocity, $v_x(t) = v_{x,0} + a_x t \rightarrow t = \frac{v_x - v_{x,0}}{a_x}$
- Position, $x(t) = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2 = x_0 + v_{x,0} \frac{v_x - v_{x,0}}{a_x} + \frac{1}{2}a_x \left(\frac{v_x - v_{x,0}}{a_x} \right)^2$

$$v_x^2 - v_{x,0}^2 = 2a_x(x - x_0)$$

Kinematics (in x-direction): constant acceleration

Position: $\vec{r}(t) = x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = \frac{dx}{dt} \quad \begin{array}{l} v_x > 0, \text{ direction } + \hat{i} \\ v_x < 0, \text{ direction } - \hat{i} \end{array}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

$$a_x = \frac{dv_x}{dt} \quad \begin{array}{l} a_x > 0, \text{ direction } + \hat{i} \\ a_x < 0, \text{ direction } - \hat{i} \end{array}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = v_{x,0} + a_x t \quad \begin{array}{l} v_x > 0, \text{ direction } + \hat{i} \\ v_x < 0, \text{ direction } - \hat{i} \end{array}$$

Position: $\vec{r}(t) = x \hat{i}$

$$x = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2 \quad \begin{array}{l} x > 0, \text{ direction } + \hat{i} \\ x < 0, \text{ direction } - \hat{i} \end{array}$$

$$v_x^2 - v_{x,0}^2 = 2a_x(x - x_0)$$

Kinematics (in y-direction): constant acceleration

Position: $\vec{r}(t) = y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = \frac{dy}{dt} \quad \begin{array}{l} v_y > 0, \text{ direction } + \hat{j} \\ v_y < 0, \text{ direction } - \hat{j} \end{array}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

$$a_y = \frac{dv_y}{dt} \quad \begin{array}{l} a_y > 0, \text{ direction } + \hat{j} \\ a_y < 0, \text{ direction } - \hat{j} \end{array}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = v_{y,0} + a_y t \quad \begin{array}{l} v_y > 0, \text{ direction } + \hat{j} \\ v_y < 0, \text{ direction } - \hat{j} \end{array}$$

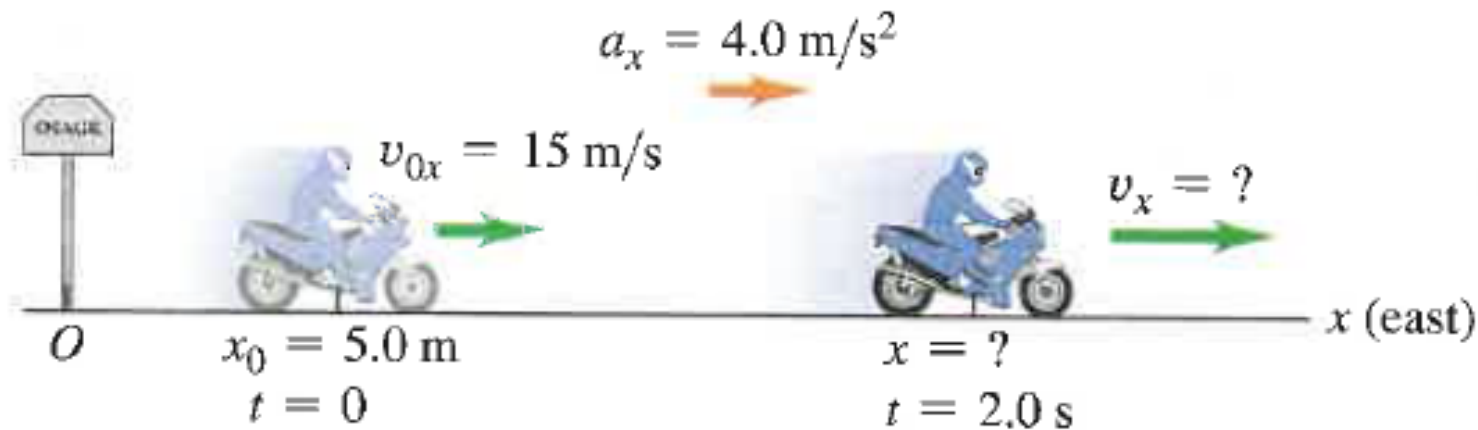
Position: $\vec{r}(t) = y \hat{j}$

$$y = y_0 + v_{y,0}t + \frac{1}{2}a_y t^2 \quad \begin{array}{l} y > 0, \text{ direction } + \hat{j} \\ y < 0, \text{ direction } - \hat{j} \end{array}$$

$$v_y^2 - v_{y,0}^2 = 2a_y(y - y_0)$$

Example:

A motorcyclist heading east through a small town accelerates at a constant 4.0 m/s^2 after he leaves the city limits signpost while he moves east at 15 m/s . (a) Find his position and velocity at $t=2.0 \text{ s}$. (b) Where is he when his speed is 25 m/s ?



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.4, p72

Example: Solution

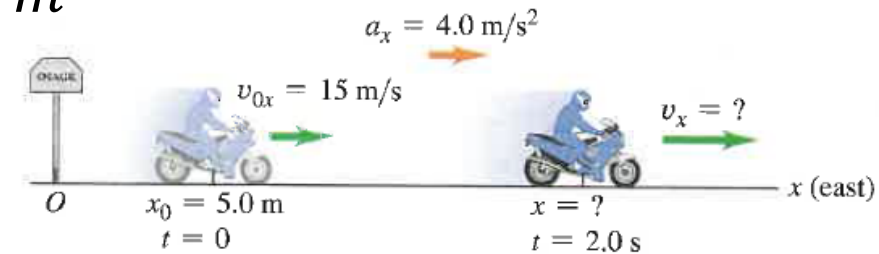
A motorcyclist heading east through a small town accelerates at a constant 4.0 m/s^2 after he leaves the city limits. At time $t=0$ he is 5.0 m east of the city-limits signpost while he moves east at 15 m/s . (a) Find his position and velocity at $t=2.0 \text{ s}$. (b) Where is he when his speed is 25 m/s ?

$$x = x_0 + v_{x,0}t + \frac{1}{2}a_xt^2 \rightarrow x = 5 + 15t + \frac{1}{2} \times 4t^2 = 43 \text{ m}$$

$$v_x = v_{x,0} + a_xt \rightarrow v_x = 15 + 4t = 23 \text{ m/s}$$

$$v_x^2 - v_{x,0}^2 = 2a_x(x - x_0) \rightarrow 25^2 - 15^2 = 2 \times 4 \times (x - 5)$$

$$\rightarrow x = 55 \text{ m}$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.4, p72

Case Problem 1:

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward speed of 15.0 m/s ; the ball is then in free fall. On its way back down, it just misses the railing. Find ~~(a) the ball's position 1.00 s and 4.00 s after leaving your hand;~~ (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; ~~(d) the ball's acceleration when it is at its maximum height.~~

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.7, p75

Case Problem 1: Solution

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward speed of 15.0 m/s; the ball is then in free fall. On its way back down, it just misses the railing. ~~Find (a) the ball's position 1.00 s and 4.00 s after leaving your hand; (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; (d) the ball's acceleration when it is at its maximum height.~~

$$v_y^2 - v_{y,0}^2 = 2a_y(y - y_0) \quad \rightarrow \quad v_y^2 - 15^2 = 2 \times (-9.8) \times (5 - 0)$$

$$\rightarrow v_y^2 = 127$$

$$\rightarrow v_y = \pm 11.3 \text{ m/s}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.7, p75

Case Problem 1: Solution

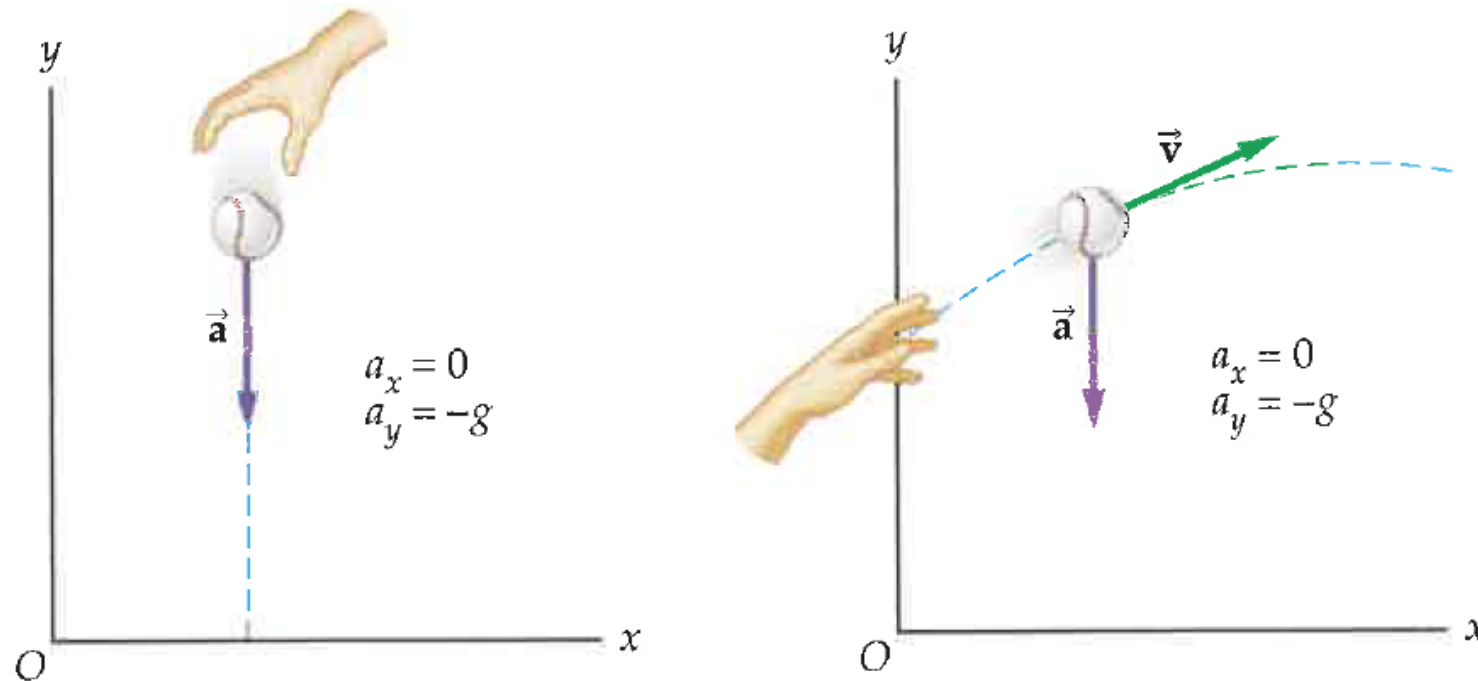
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$$v_y^2 - v_{y,0}^2 = 2a_y(y - y_0) \quad \rightarrow \quad 0^2 - 15^2 = 2 \times (-9.8) \times (y - 0)$$

$$\rightarrow y = 11.5 \text{ m}$$

Projectile Motion

Projectile motion describes the motion of an object thrown into air, experiencing weight only. Air resistance is ignored.



▲ **FIGURE 4-2 Acceleration in free fall**

Kinematics (Projectile motion)

Position: $\vec{r}(t) = x \hat{i} + y \hat{j}$

Velocity: $\vec{v}(t) = v_x \hat{i} + v_y \hat{j}$

$$v_x = \frac{dx}{dt} \quad v_y = \frac{dy}{dt}$$

Acceleration:

$$\vec{a}(t) = a_x \hat{i} + a_y \hat{j} = -g \hat{j}$$

$$a_x = 0$$

$$a_y = -g$$

Acceleration: $\vec{a}(t) = a_y \hat{j} = -g \hat{j}$

Velocity: $\vec{v}(t) = v_x \hat{i} + v_y \hat{j}$

$$v_x(t) = v_{x,0}$$

$$v_y(t) = v_{y,0} - gt$$

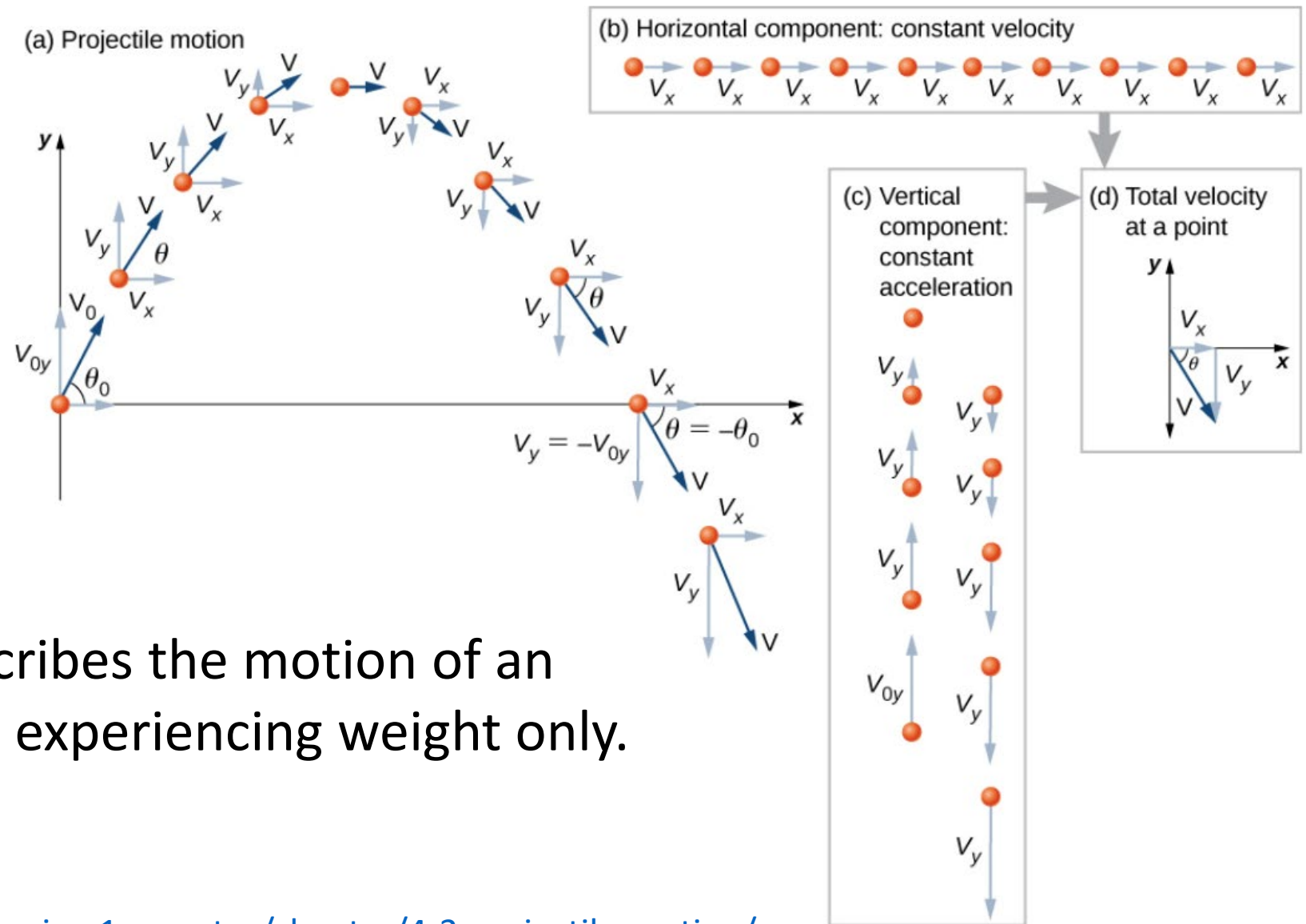
Position: $\vec{r}(t) = x \hat{i} + y \hat{j}$

$$x = x_0 + v_{x,0}t$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$v_y^2 - v_{y,0}^2 = -2g(y - y_0)$$

Projectile Motion



Projectile motion describes the motion of an object thrown into air, experiencing weight only.

<https://opentextbc.ca/universityphysicsv1openstax/chapter/4-3-projectile-motion/>

Projectile Motion: Zero initial velocity

1-D motion in y direction



$$\text{Acceleration: } \vec{a}(t) = a_y \hat{j} = -g \hat{j}$$

$$\text{Velocity: } \vec{v}(t) = v_x \hat{i} + v_y \hat{j}$$

$$\cancel{v_x(t) = v_{x,0} = 0}$$

$$v_y(t) = v_{y,0} - gt$$

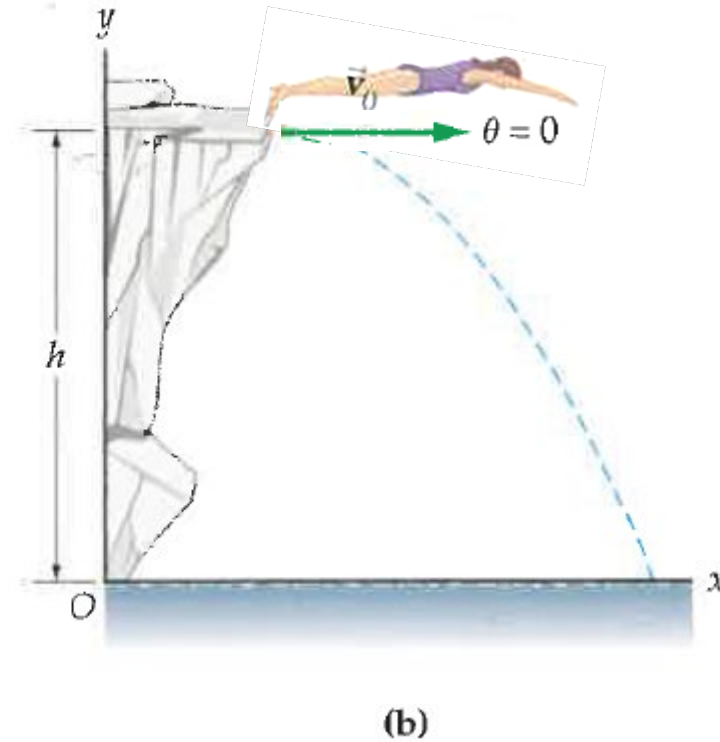
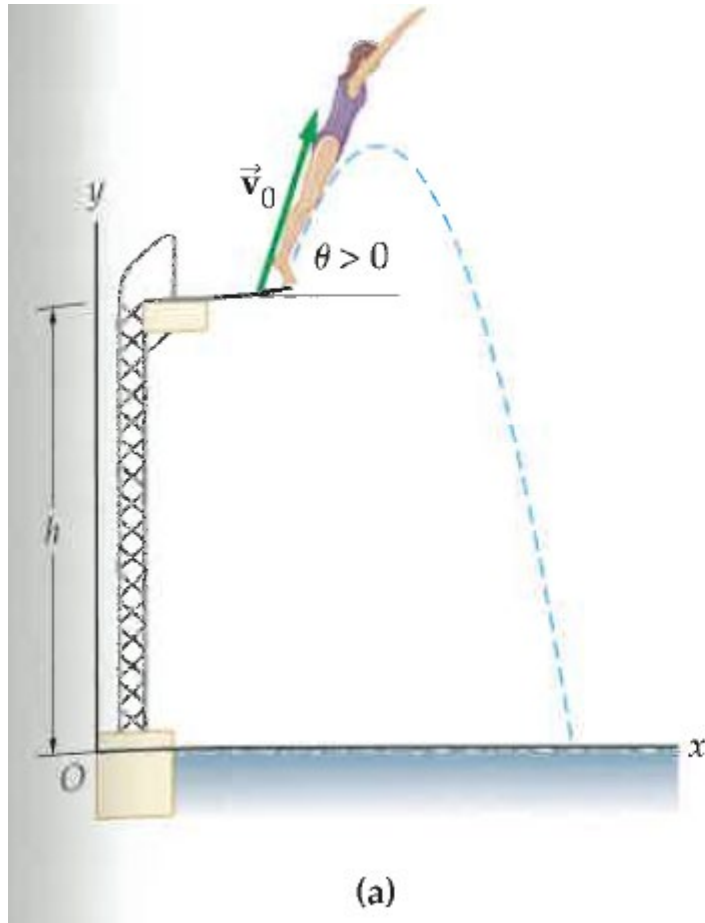
$$\text{Position: } \vec{r}(t) = x \hat{i} + y \hat{j}$$

$$\cancel{x = x_0 + v_{x,0}t = 0}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$v_y^2 - v_{y,0}^2 = -2g(y - y_0)$$

Projectile Motion: Launch angle



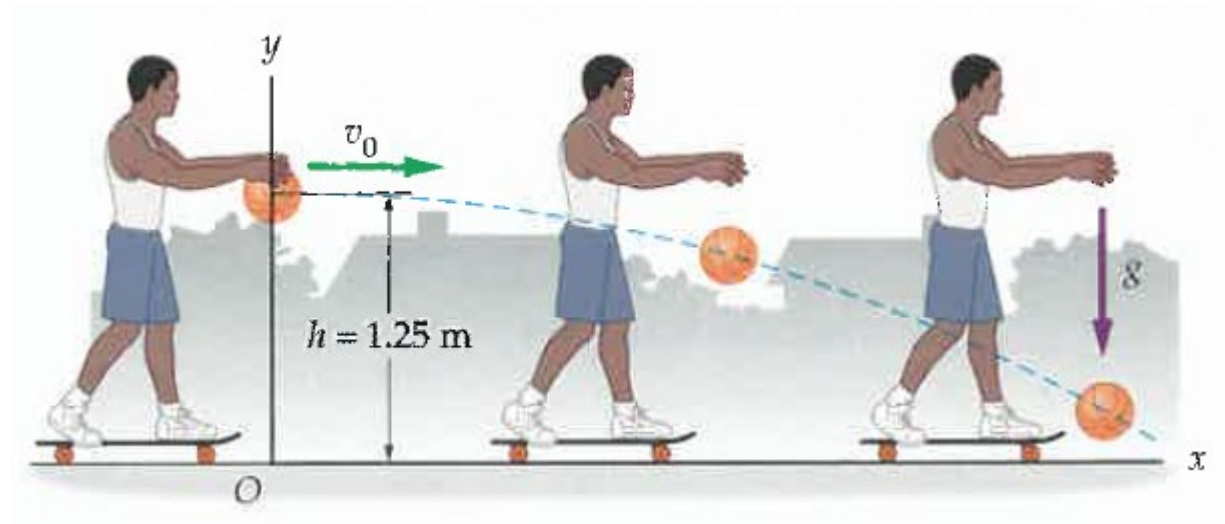
J. S. Walker, Physics: p85

Projectile Motion: Zero launch angle

When a projectile is launched with an initial velocity $\vec{v}_0$ horizontally,

$$v_{x,0} = v_0$$

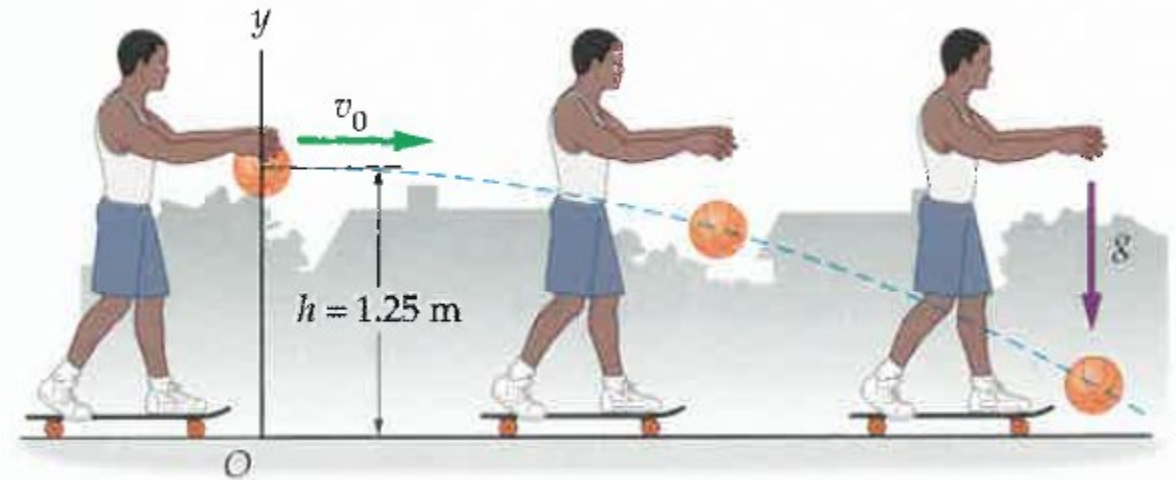
$$v_{y,0} = 0$$



J. S. Walker, Physics: Example 4-3, p85

Example

A person skateboarding with a constant speed of 1.3 m/s releases a ball from a height of 1.25 m above the ground. Given that $x_0 = 0$ and $y_0 = h = 1.25 \text{ m}$, (a) find x and y at $t = 0.25 \text{ s}$ and (b) find the velocity, speed, and direction of motion of the ball at $t = 0.5 \text{ s}$. Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics: Example 4-3, p85

Example: Solution

A person skateboarding with a constant speed of 1.3 m/s releases a ball from a height of 1.25m above the ground. Given that $x_0 = 0$ and $y_0 = h = 1.25 \text{ m}$, **(a) find x and y at $t = 0.25 \text{ s}$** and (b) find the velocity, speed, and direction of motion of the ball at $t = 0.5 \text{ s}$. Take $g = 9.81 \text{ m/s}^2$.

$$x = x_0 + v_{x,0}t \quad \rightarrow \quad x(0.25) = 0 + 1.3 \times 0.25 = 0.325\text{m}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \quad \rightarrow \quad y(0.25) = 1.25 + 0 - \frac{1}{2}9.81 \times (0.25)^2 = 0.943\text{m}$$

Example: Solution

A person skateboarding with a constant speed of 1.3 m/s releases a ball from a height of 1.25m above the ground. Given that $x_0 = 0$ and $y_0 = h = 1.25 \text{ m}$, (a) find x and y at $t = 0.25 \text{ s}$ and **(b) find the velocity, speed, and direction of motion of the ball at $t = 0.5 \text{ s}$.** Take $g = 9.81 \text{ m/s}^2$.

$$v_x(t) = v_{x,0} \rightarrow v_x(0.5) = 1.3 \text{ m/s}$$

$$v_y(t) = v_{y,0} - gt \rightarrow v_y(0.5) = 0 - 9.81 \times 0.5 = -4.91 \text{ m/s}$$

$$\vec{v}(t) = v_x \hat{i} + v_y \hat{j} \rightarrow \vec{v}(0.5) = (1.3 \hat{i} - 4.91 \hat{j}) \text{ m/s}$$

$$v(0.5) = \sqrt{(1.3)^2 + (-4.91)^2} = 5.08 \text{ m/s}$$

$$\theta = \tan^{-1} \frac{v_y(0.5)}{v_x(0.5)} = \tan^{-1} \frac{-4.91}{1.3} = 360^\circ - 75.2^\circ = 284.8^\circ$$

J. S. Walker, Physics: Example 4-3, p85

Case Problem 2

A motorcycle stunt rider rides off the edge of a cliff. Just at the edge his velocity is horizontal, with magnitude 9.0 m/s . Find the motorcycle's position from the edge of the cliff, and velocity 0.50 s after it leaves the edge of the cliff. Take $g = 9.8 \text{ m/s}^2$.

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.6, p102

Case Problem 2: Solution

A motorcycle stunt rider rides off the edge of a cliff. Just at the edge his velocity is horizontal, with magnitude 9.0 m/s. Find the motorcycle's **position** from the edge of the cliff, and velocity 0.50 s after it leaves the edge of the cliff. Take $g = 9.8 \text{ m/s}^2$.

$$x = x_0 + v_{x,0}t \rightarrow x(0.5) = 9.0 \times 0.5 = 4.5 \text{ m}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \rightarrow y(0.5) = 0 - \frac{1}{2} \times 9.8 \times (0.5)^2 = -1.2 \text{ m}$$

$$\vec{x}(t) = x\hat{i} + y\hat{j} \rightarrow \vec{x}(0.5) = (4.5\hat{i} - 1.2\hat{j}) \text{ m}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.6, p102

Case Problem 2: Solution

A motorcycle stunt rider rides off the edge of a cliff. Just at the edge his velocity is horizontal, with magnitude 9.0 m/s. Find the motorcycle's position from the edge of the cliff, and **velocity** 0.50 s after it leaves the edge of the cliff. Take $g = 9.8 \text{ m/s}^2$.

$$v_x(t) = v_{x,0} \rightarrow v_x(0.5) = 9.0 \text{ m/s}$$

$$v_y(t) = v_{y,0} - gt \rightarrow v_y(0.5) = 0 - 9.8 \times 0.5 = -4.9 \text{ m/s}$$

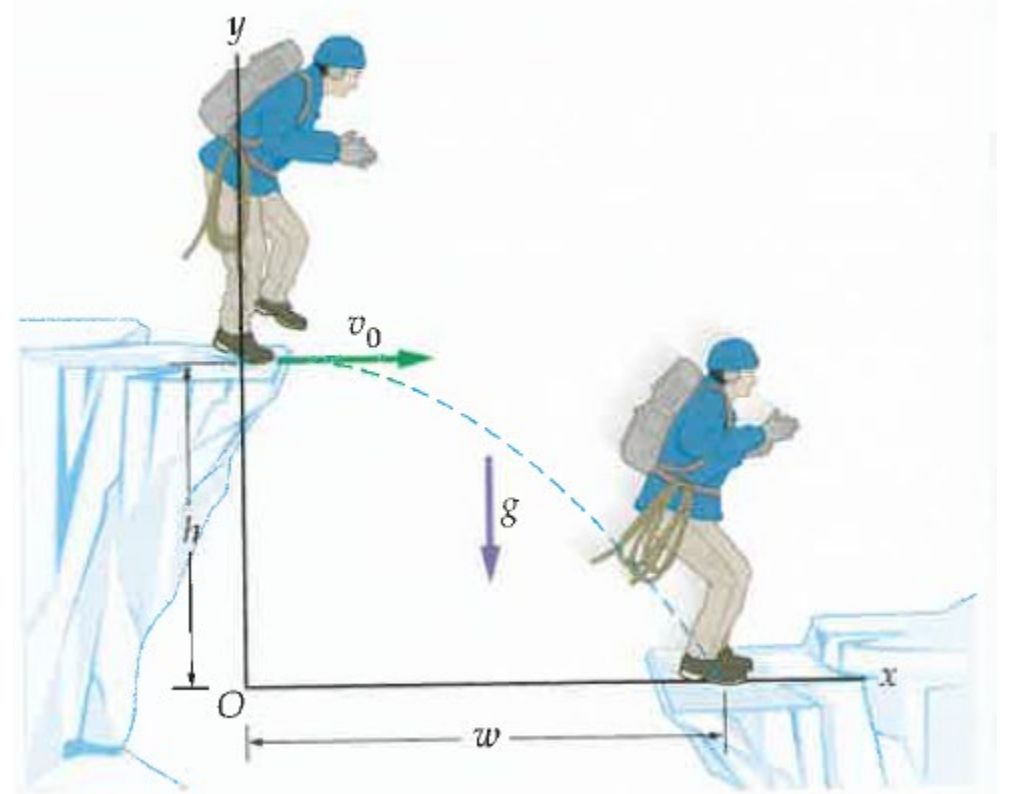
$$\vec{v}(t) = v_x \hat{i} + v_y \hat{j} \rightarrow \vec{v}(0.5) = (9.0 \hat{i} - 4.9 \hat{j}) \text{ m/s}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.6, p102

Example

A mountain climber encounters a crevasse in an ice field. The opposite side of the crevasse is 2.75m lower, and is separated horizontally by a distance of 4.10m. To cross the crevasse, the climber gets a running start and jumps in the horizontal direction. Take $g = 9.81 \text{ m/s}^2$.

- a) What is the minimum speed needed by the climber to safely cross the crevasse?
- b) If, instead, the climber's speed is 6.00m/s, where does the climber land, and what is the climber's speed on landing?



Example: Solution

a)

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \rightarrow 0 = 2.75 - \frac{1}{2}gt^2 \rightarrow t = \sqrt{\frac{5.5}{g}}$$

$$x = x_0 + v_{x,0}t \rightarrow 4.10 = v_{x,0} \sqrt{\frac{5.5}{g}} \rightarrow v_{x,0} = 4.10 \times \sqrt{\frac{g}{5.5}} = 5.48 \text{ m/s}$$

Example: Solution

$$\text{b)} \quad y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \quad \rightarrow \quad 0 = 2.75 - \frac{1}{2}gt^2 \quad \rightarrow \quad t = \sqrt{\frac{5.5}{g}}$$

$$x = x_0 + v_{x,0}t \quad \rightarrow \quad x = v_{x,0} \sqrt{\frac{5.5}{g}} = 6.00 \times \sqrt{\frac{5.5}{g}} = 4.49 \text{ m}$$

Example: Solution

$$\text{b)} \quad y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \rightarrow 0 = 2.75 - \frac{1}{2}gt^2 \rightarrow t = \sqrt{\frac{5.5}{g}}$$

$$v_y(t) = v_{y,0} - gt \rightarrow v_y = -g \sqrt{\frac{5.5}{g}} = -7.35 \text{ m/s}$$

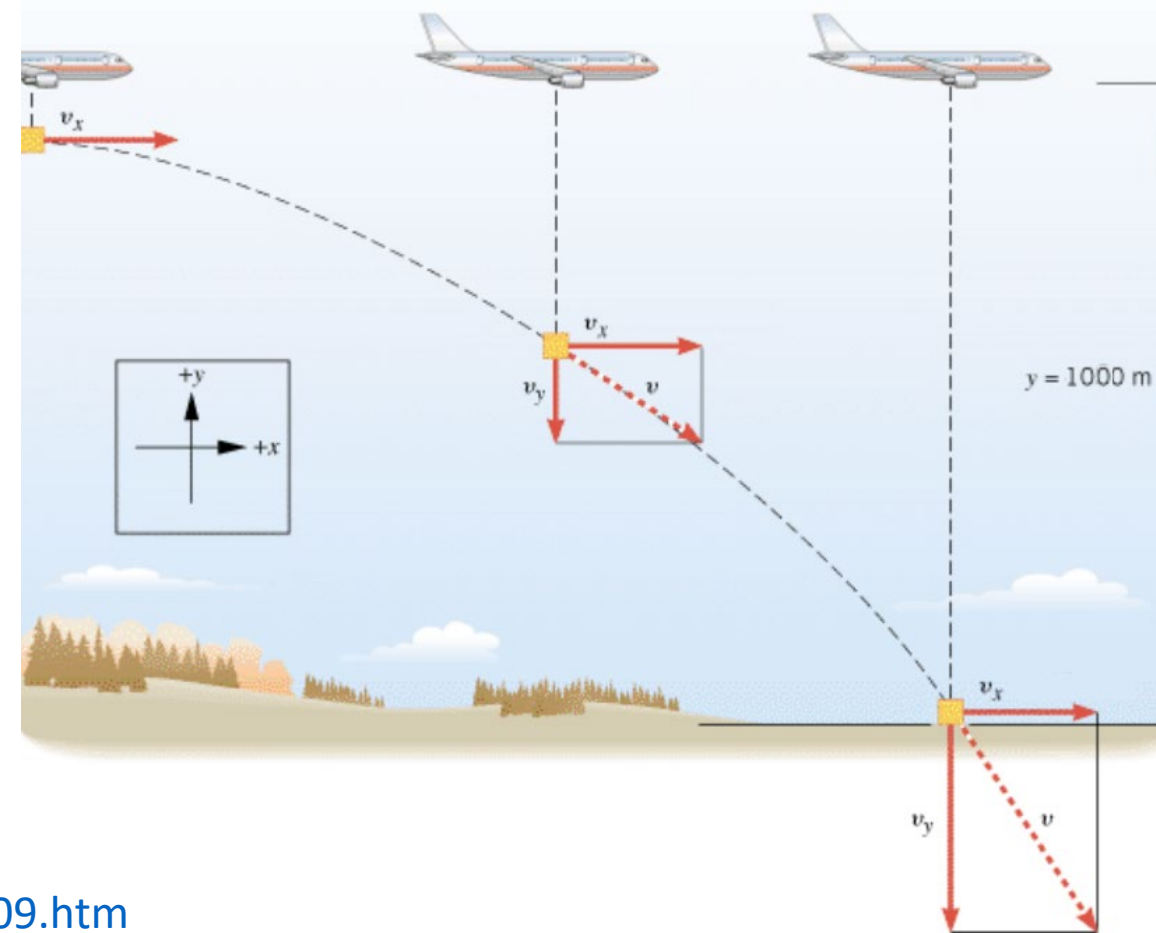
$$\vec{v} = v_x \hat{i} + v_y \hat{j} = 6.00 \hat{i} - 7.35 \hat{j}$$

$$v = \sqrt{(v_x)^2 + (v_y)^2} = \sqrt{(6.00)^2 + (-7.35)^2} = 9.49 \text{ m/s}$$

Case Problem 3

A plane is flying at a constant velocity of 100 m/s horizontally at an altitude of 1000 m. It delivers a package of goods to land on a target area on the ground. Take $g = 10 \text{ m/s}^2$.

- a) What is the time required for the package to land on the ground?
- b) How far before the target area should the plane release the package?
- c) What is the velocity of the package when it lands on the ground?



<https://www.webassign.net/ebooks/cj6demo/xlinks/c03-sec1-0009.htm>

Case Problem 3: Solution

$$\text{a)} \quad y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 1000 + 0t - \frac{1}{2}gt^2$$

$$1000 + 0T - \frac{1}{2}gT^2 = 0 \rightarrow T = \sqrt{\frac{2000}{g}} = 14.1 \text{ s}$$

$$\text{b)} \quad x = x_0 + v_{x,0}t = 0 + 100t$$

$$x(T) = 0 + 100T = 1410 \text{ m}$$

Case Problem 3: Solution

$$\text{c)} \quad v_y = v_{y0} - gt = 0 - gt$$

$$v_y(T) = 0 - gT = -141 \text{ m/s}$$

$$v_x(T) = v_{x,0} = 100 \text{ m/s}$$

$$\vec{v}(T) = v_x(T)\hat{i} + v_y(T)\hat{j}$$

Off-class practice question(s)

Case Problem:

Jets at an airport accelerate from rest at one end of a runway, and must attain takeoff speed before reaching the other end of the runway. A plane has acceleration a_x and takeoff speed v_x . Find the minimum length of runway required.

Case Problem: Solution

Jets at an airport accelerate from rest at one end of a runway, and must attain takeoff speed before reaching the other end of the runway. A plane has acceleration a_x and takeoff speed v_x . Find the minimum length of runway required.

$$v_x^2 - v_{x,0}^2 = 2a_x(x - x_0)$$

$$v_x^2 = 2a_x(x - x_0)$$

$$x - x_0 = \frac{v_x^2}{2a_x} \rightarrow l \geq x - x_0 \rightarrow l \geq \frac{v_x^2}{2a_x}$$

Case Problem

A car is moving along x direction. The driver suddenly found a roadblock ahead, which is at a distance of l m from the car. At that moment, the velocity of the car is v_0 m/s. The driver immediately hit the brake to make the car decelerate with constant deceleration $-a$ m/s² and finally stopped the car before the roadblock. Find the minimum value of a required in terms of v_0 and l .

Case Problem: Solution

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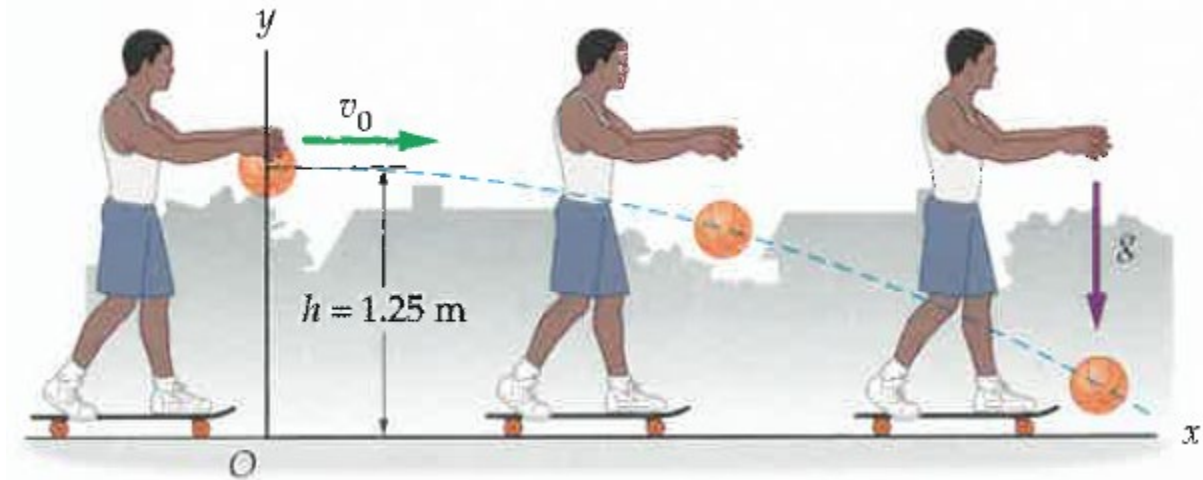
$$v_x^2 - v_{x,0}^2 = 2a_x(x - x_0)$$

$$0 - v_0^2 = 2(-a)(x - x_0)$$

$$x - x_0 = \frac{v_0^2}{2a} \quad \rightarrow \quad l \geq x - x_0 \rightarrow \quad l \geq \frac{v_0^2}{2a} \rightarrow \quad a \geq \frac{v_0^2}{2l}$$

Case Problem

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J. S. Walker, Physics: Example 4-3, p85

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$$x = x_0 + v_{x,0}t \quad \rightarrow \quad x(0.5) = 0 + 1.3 \times 0.5 = 0.65m$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \quad \rightarrow \quad y(0.5) = 1.25 + 0 - \frac{1}{2}9.81 \times (0.5)^2 = 0.0238m$$